A brief Introduction To GIS



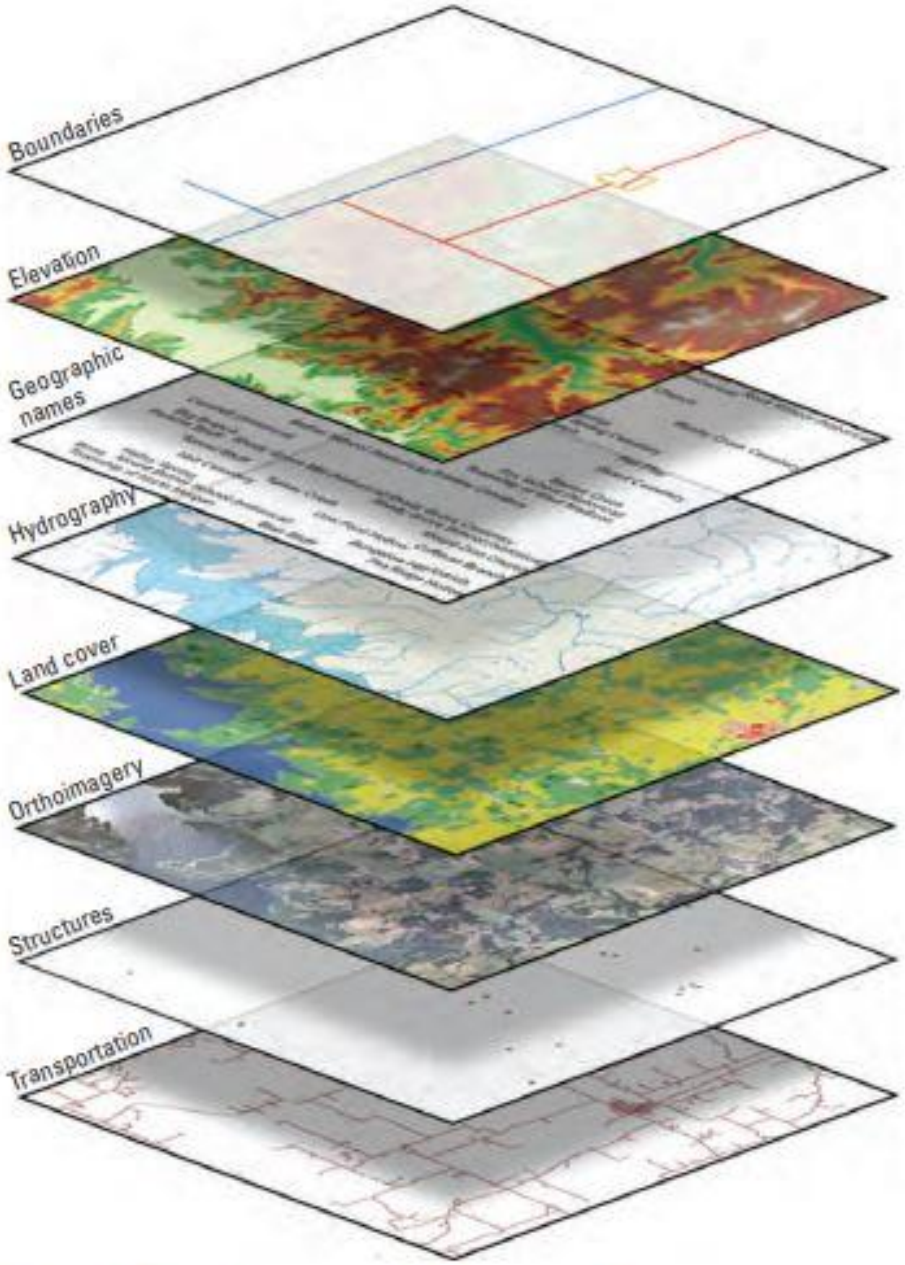


Figure 1. Eight base layers of *The National Map*.

What is GIS?
Geographic Information System

"..integrated computer hardware and software that store, manage, analyze, edit, output, and visualize geographic data. Much of this often happens within a spatial database…In a broader sense, one may consider such a system also to include human users and support staff, procedures and workflows, the body of knowledge of relevant concepts and methods, and institutional organizations."-Wikipedia

"Any system for capturing, storing, checking, and displaying data related to positions on the Earth's surface." -National Geographic Education Encyclopedia

In a GIS, you connect ***data*** with ***geography***

-GISgeography.com

Two popular GIS programs



- Expensive, commercial
- Windows (but there are work-arounds)
- More stable, consistent, compatible; less flexible, less customizable
- Cannot open some QGIS files
- Until last year, it was the ONLY way to run LAHARZ!



- Free, open source
- Windows, Mac, Linux, Android
- More flexible, customizable (plug-ins, scripting), but can have bugs
- Can open ArcGIS formatted files
- Thanks to Keith Blair, there is now an open-source, stand-alone python version of LAHARZ!

Two popular GIS programs





- For this year's course, we will use QGIS and the open-source version of LAHARZ on the VICTOR platform
- However, the ArcGIS course materials (ArcMap and ArcPro) and LAHARZ code are still available and will be provided to you in the class files

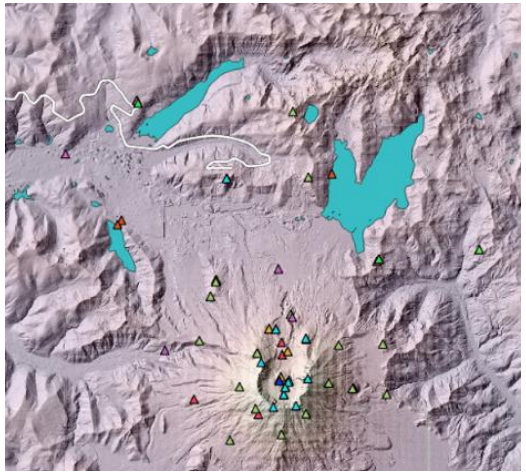
Representing Geographic Features

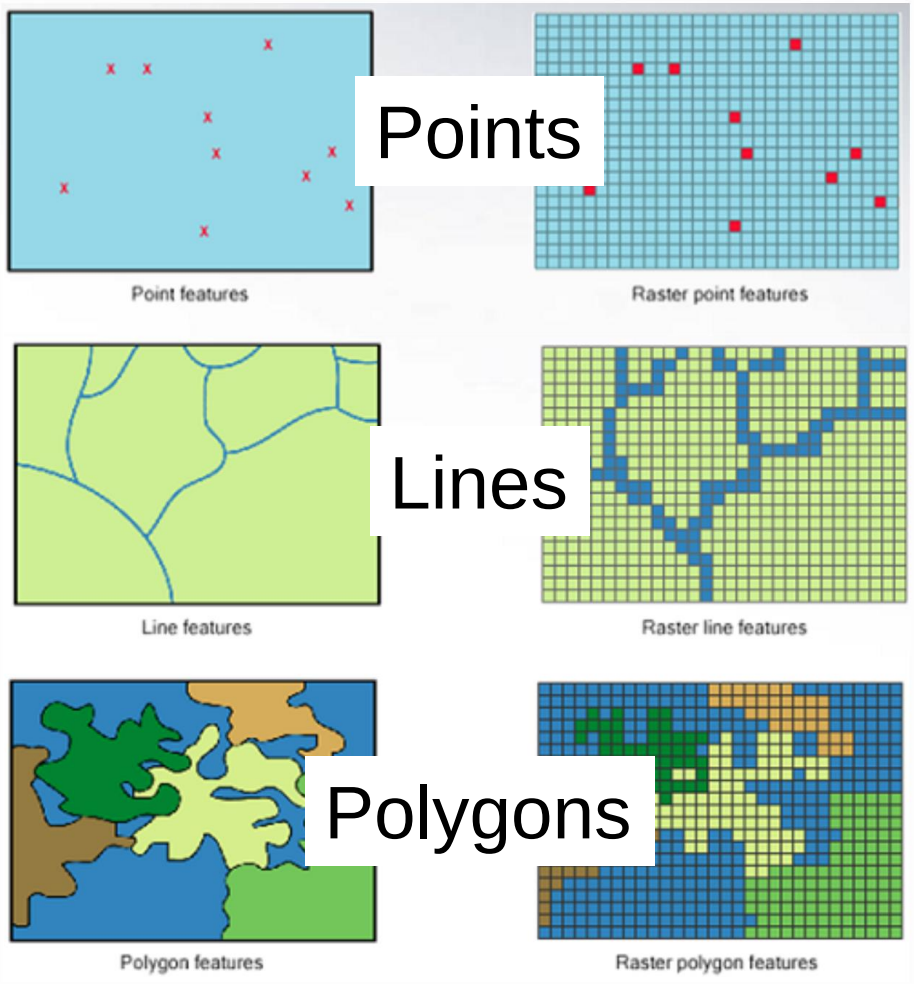
Vector Data

Raster Data

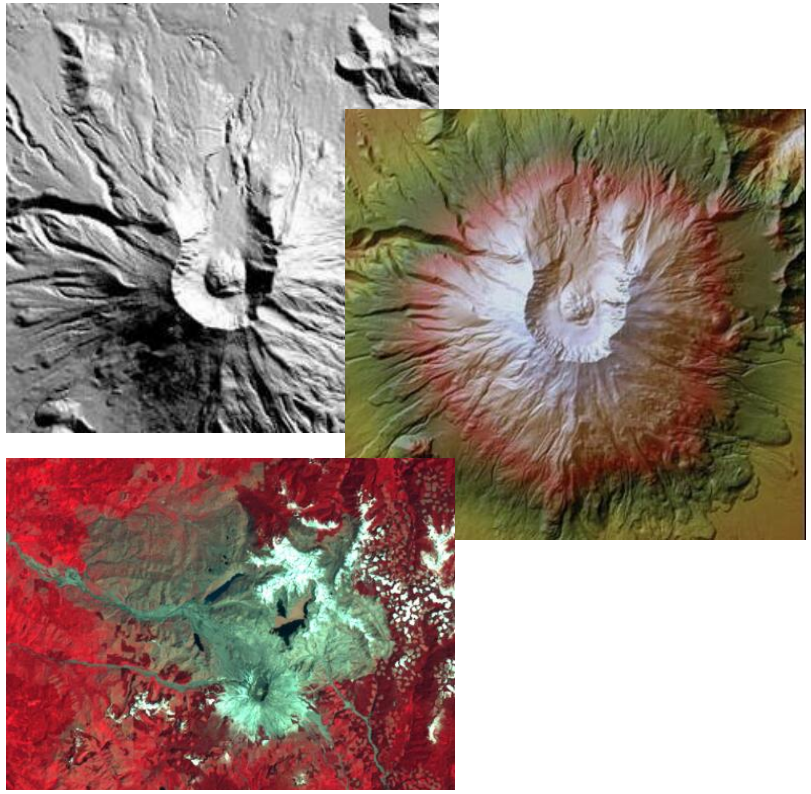
"Raster is faster; Vector is correct-er"

Features like road networks, lakes, park boundaries, station locations, etc. are usually vectors (in ArcGIS these are shapefiles)





Digital elevation models (DEMs), hillshades, satellite images are all rasters



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GIS works when you combine geographic location with attribute information

Where is it?

Spatial Coordinates
(204654, 3984988)

+

What is it?

Attributes

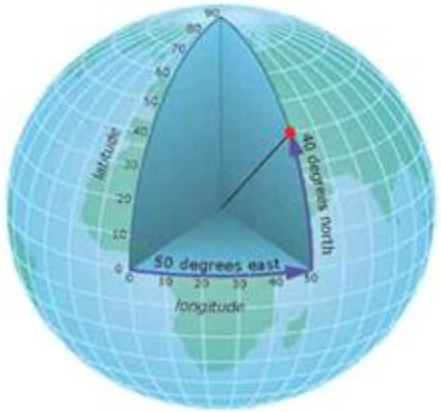
FID	Shape *	Volcano	Instrument	Solar?
024	point	Kilauea	Seismometer	Yes

and you display it with scale, direction, and system of measurement

Overview: Coordinate system types

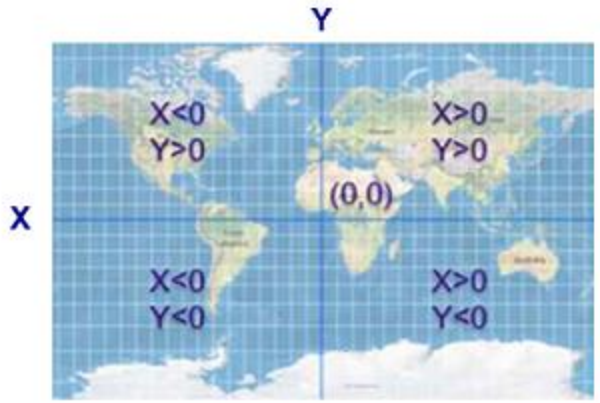
Geographic Coordinate Systems (GCS)

- Defines ***where*** *the data is located on the earth*
  - Angular measurements
  - Latitude & Longitude
  - Major variable: **datum**



Projected Coordinate Systems (PCS)

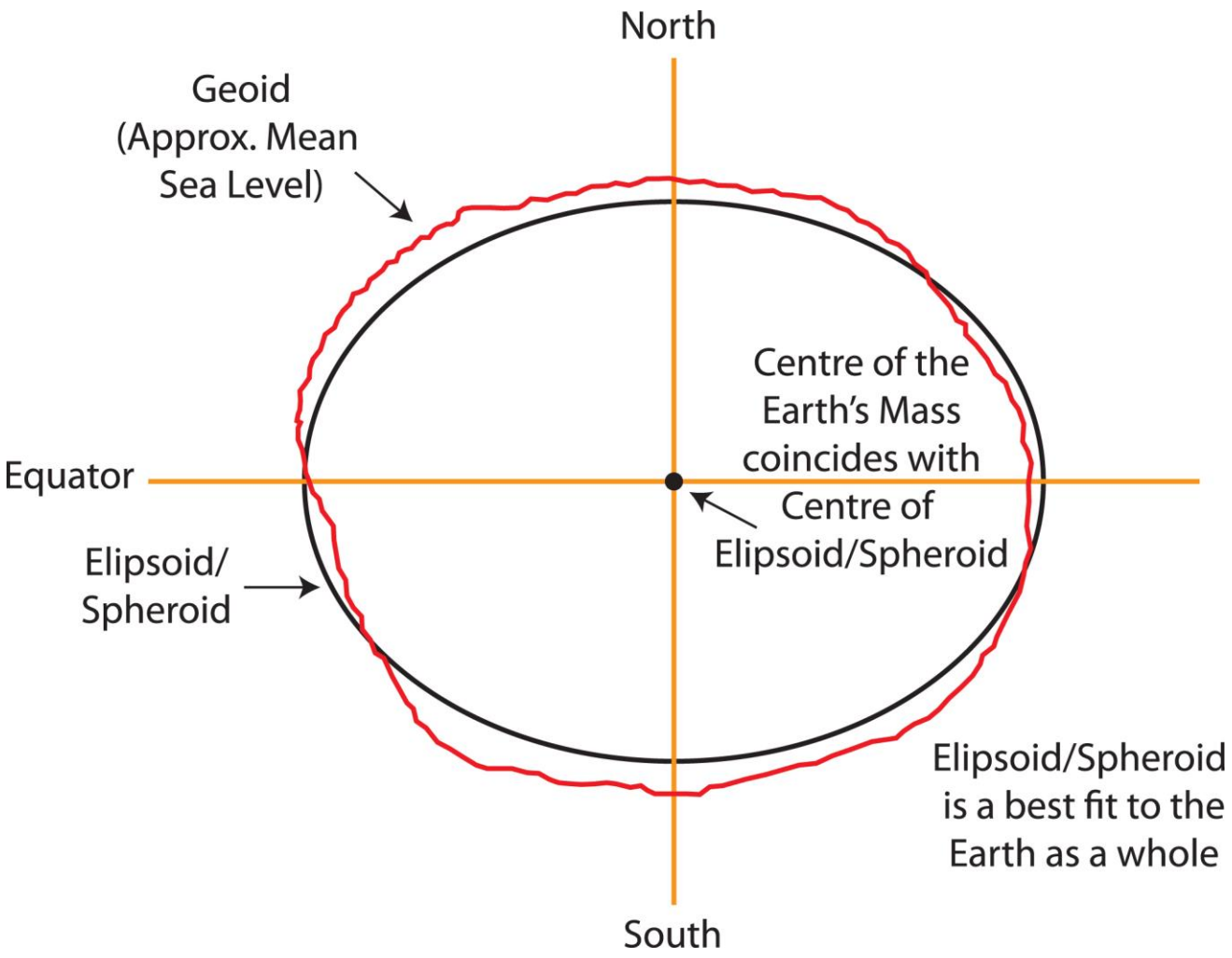
- Tells the data ***how*** *to draw on a flat surface*
  - Linear measurements
  - Feet & Meters
  - Major variable: **flattening method**



Projections can preserve size, shape, and distance, but not all three at the same time!

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Modelling the earth's shape



- Geoid
  - approximates mean sea level without winds or tides
- Spheroid (ellipsoid)
  - Mathematically idealized approximation of the size and shape of Earth
- WGS84 datum
  - World Geodetic System
  - Defines an earth-centered coordinate system with the origin coordinate at the Earth's center of mass
  - Used by GPS and satellite data

https://en.wikipedia.org/wiki/World_Geodetic_System

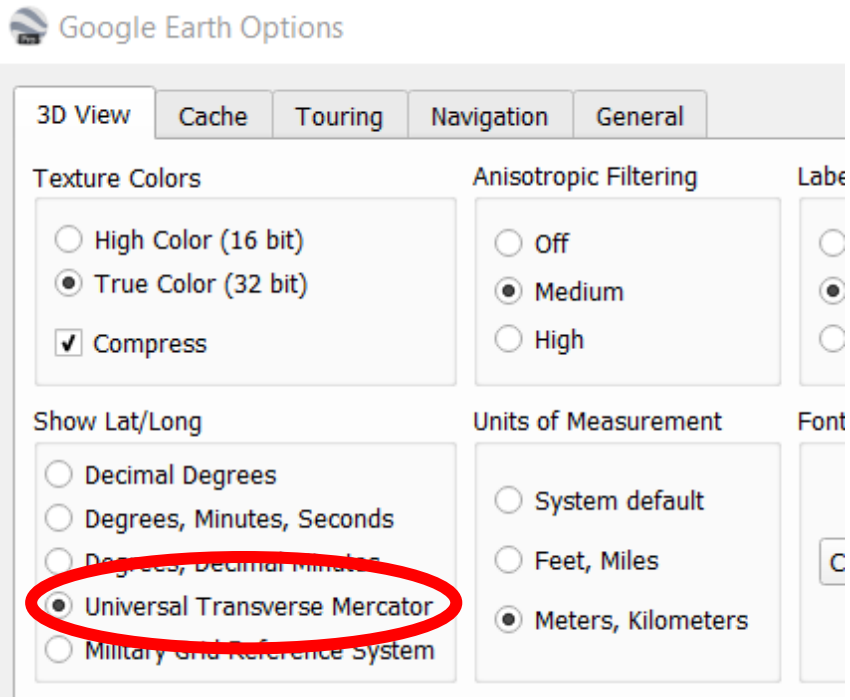
Universal Transverse Mercator (UTM)

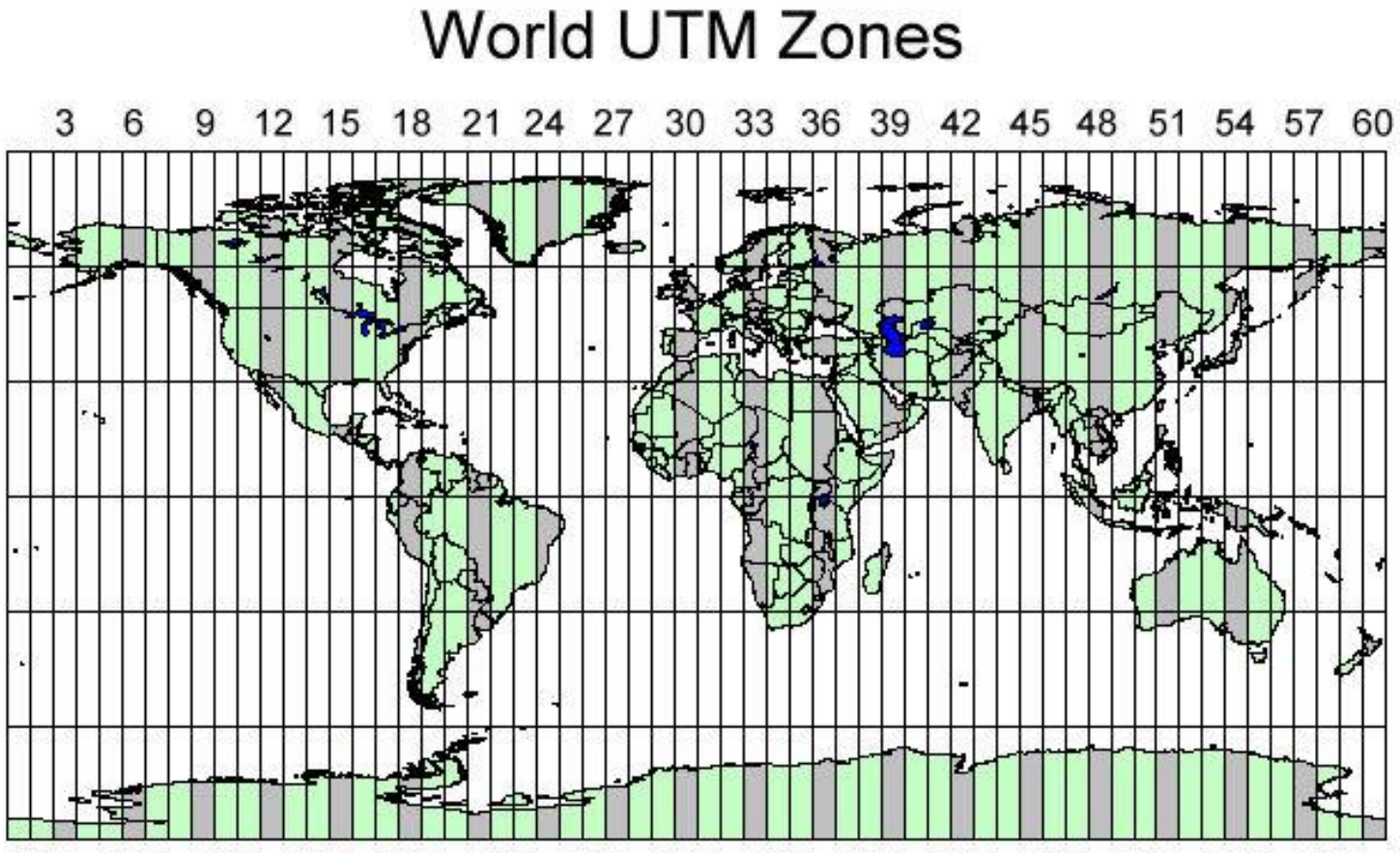
- Divides earth into 60 zones and projects each to the plane as a basis for its coordinates
- Specifying a location means specifying the zone and the *x*, *y* coordinate in that plane.
- Uses a grid-based, cartesian system so calculating distances is easy (which is why we use it for ArcGIS versions of LAHARZ)

https://hub.arcgis.com/datasets/esri::world-utm-grid/explore?location

- For the ArcMap and ArcPro versions of LAHARZ, the DEM needs to be in UTM
- For the VICTOR and stand-alone python versions, projections are handled internally

Google Earth: tools/options





SRTM

Space Shuttle Radar Topography Mission Survey
Collected for 11 days in February 2000
Radar “sees” through clouds
60° N to 56° S, free
X-band ~6cm wavelength
30-meter global dataset
Version 3 “MEASURES” released 2020



TerraSAR-X & Tandem-X

Pair of satellites flying in close formation
Collected over 6 years of 2010-2015
Radar “sees” through clouds
0° N to 0° S, free
X-band ~6cm wavelength
90-meter global dataset
(future release of 30 or 12 meters?)



ASTER

Advanced Spaceborne Thermal Emission and Reflection Radiometer (optical instrument) on Terra
4 million images 2000-2019
Data gaps: persistent clouds; filled with SRTM
83° N to 83° S, free
Optical stereo derived; elevations may include vegetation
30-meter global dataset
Version 3 released 2019





Precise, not accurate



Precise and accurate



Not precise, not accurate



Accurate, not precise

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How much did my phone cost?

	Precise	Not Precise
Not Accurate		
Accurate		

$400

$399.99

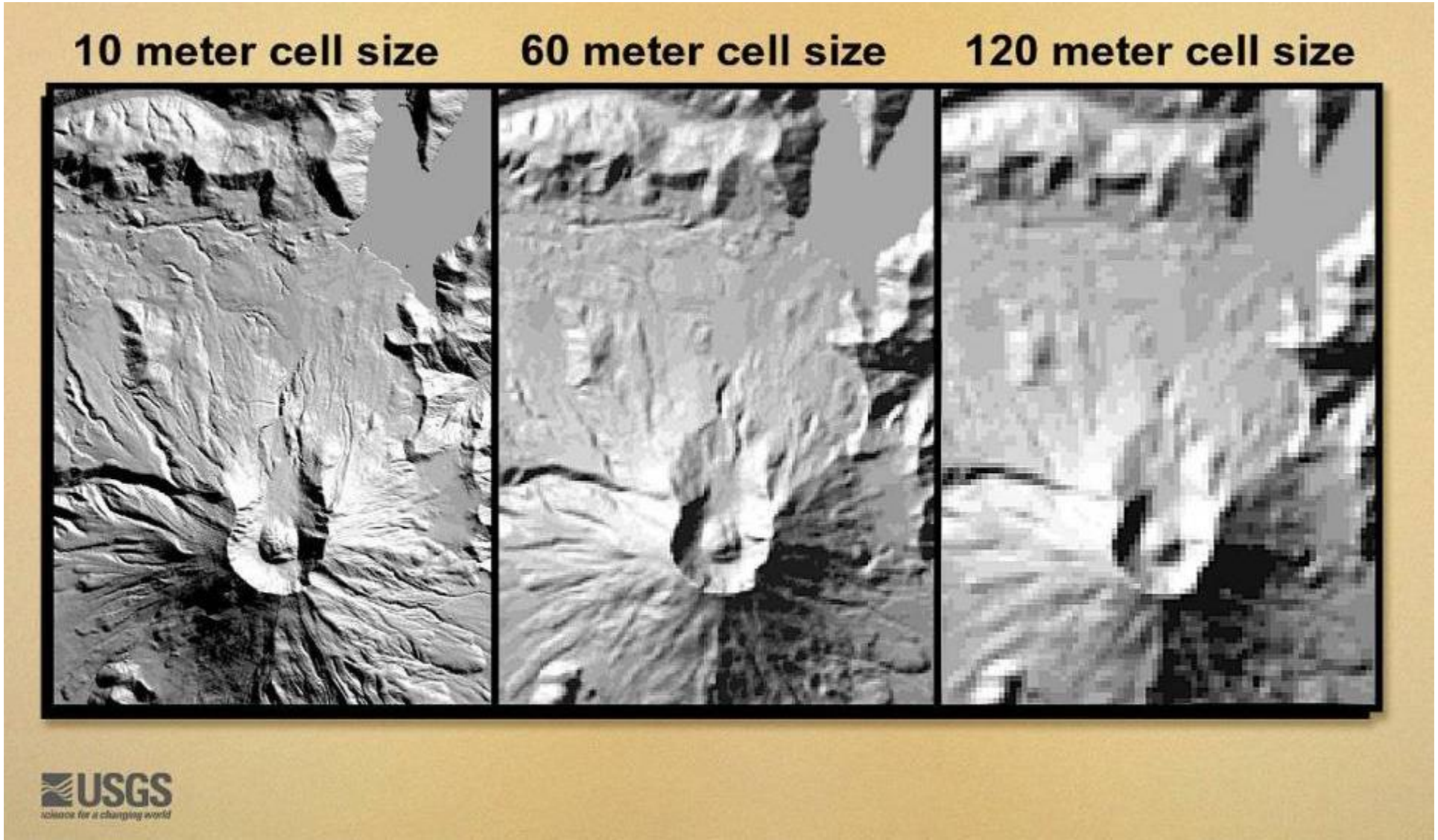
$3,999,999.999

$4,000,000

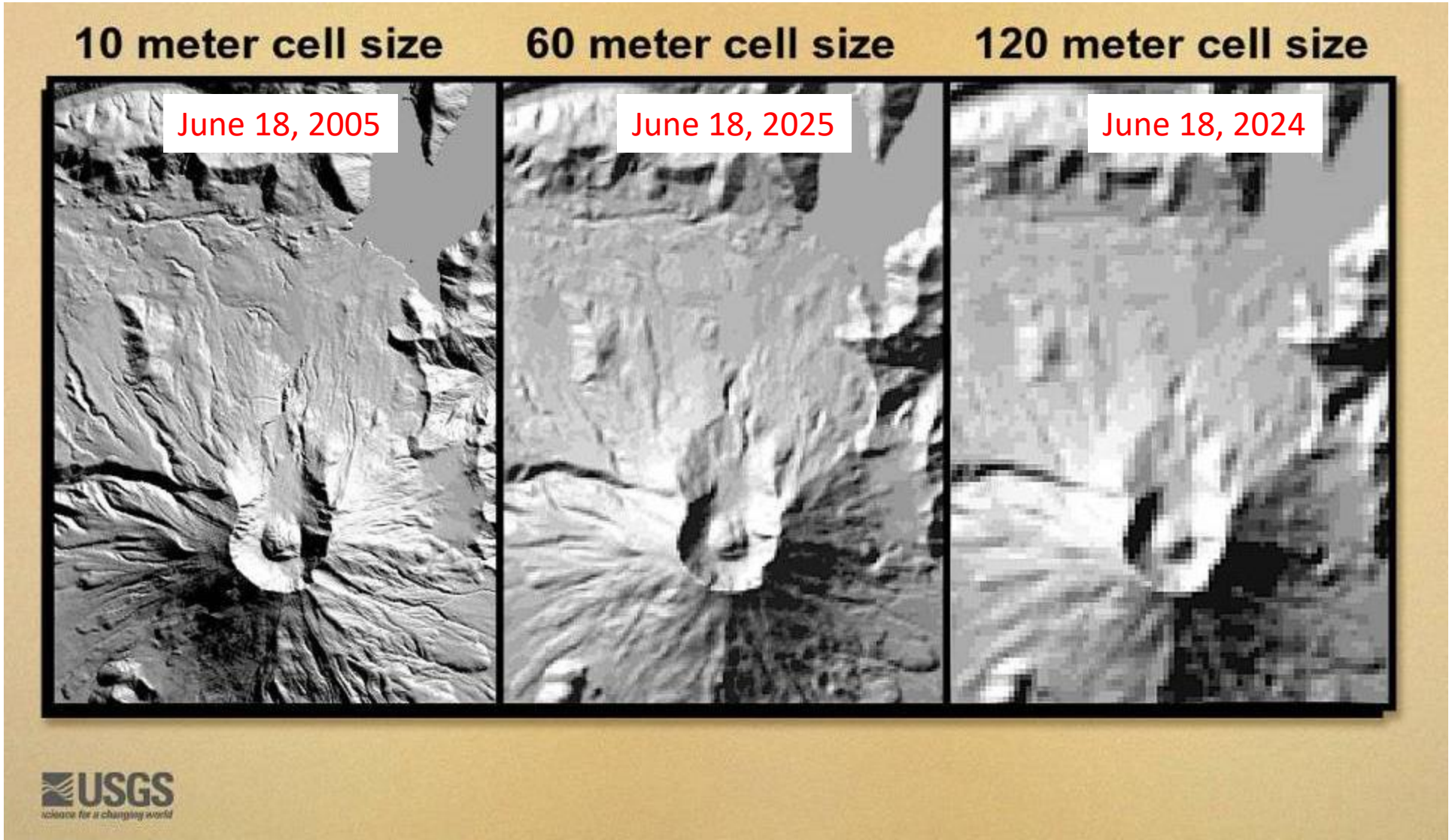
Precision ≠ Accuracy

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Which DEM would you want to use?



Which DEM would you want to use?



DEMs on VICTOR



OpenTopography

High-Resolution Topography Data and Tools

		ArcticDEM 2m
4	⊞ Global Ecosystem Dynamics Investigation (GEDI) L3 Gridded Land Surface Metrics Open Data	GEDI L3
5	⊞ Continental Europe Digital Terrain Model Open Data	Europe DTM
6	⊞ Copernicus Global Digital Elevation Models Open Data	Copernicus 30m Copernicus 90m
7	⊞ NASADEM Global Digital Elevation Model Open Data	NASADEM
8	⊞ Global Bathymetry and Topography at 15 Arc Sec: SRTM15+ V2.5.5 Open Data	SRTM15+
9	⊞ Global Multi-Resolution Topography (GMRT) Data Synthesis Open Data	GMRT
10	⊞ ALOS World 3D - 30m Open Data	AW3D30 AW3D30 Ellipsoidal
11	⊞ Shuttle Radar Topography Mission (SRTM) Global Open Data	SRTM GL3 SRTM GL1 SRTM GL1 Ellipsoidal

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Using VICTOR to get a DEM

